

Molecular dynamic simulations of heterophilic antibody binding to diagnostic IgG antibodies: towards understanding sample interference in rapid immunoassays

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Abstract. Coarse-grained molecular dynamics study of low-affinity, polyspecific antibodies in miniaturized, microfluidic, quantitative rapid diagnostics.

Keywords: Microfluidic diagnostics; Matrix Interference; Heterophilic Antibodies; Molecular Dynamics; Computer simulations

1 Introduction and state-of-the-art

Point-of-care (POC) diagnostics enable decentralized testing, yet most current technologies remain qualitative. Quantitative capabilities are essential for disease monitoring and early diagnosis. Protein biomarkers, whose concentrations in biological fluids are closely linked to these conditions, are therefore critical targets, making their accurate quantification key for diagnosis, prognosis, and monitoring of therapeutic response [1], [2], [3], [4], [5].

Microfluidic diagnostics have emerged as a key technology for rapid quantification in POC applications due to their ability to precisely manipulate small fluid volumes, reducing reagent consumption and significantly accelerating reaction kinetics [6]. These systems enhance antigen–antibody interactions, enabling faster and more sensitive assays. Several microfluidic platforms have been developed, including lab-on-a-chip systems, droplet-based microfluidics, and paper-based devices [7]. Among these, microcapillary film (MCF) technology has gained attention as a versatile platform composed of parallel microcapillaries that provide a high surface area for antibody immobilization and support

efficient, high-throughput immunoassays under controlled flow conditions [8].

Immunoassays rely on the highly specific interaction between antibodies and antigens, enabling the detection of target proteins at very low concentrations [9].

Immunoassays can be broadly classified as homogeneous or heterogeneous depending on whether separation steps are required during the analytical process. In homogeneous immunoassays, detection occurs directly in solution without the need to remove unbound reagents. In contrast, heterogeneous immunoassays involve separation or washing steps to remove unbound molecules and are frequently implemented on solid-phase platforms [9].

Several immunoassay formats have been developed depending on the characteristics of the analyte and the intended application, including sandwich, competitive, and indirect assays. In sandwich immunoassays, the analyte is captured between two antibodies that recognize different epitopes on the same molecule, typically providing high sensitivity and specificity. In contrast, competitive immunoassays rely on the competition between labeled and unlabeled analytes for the same antibody binding sites and are particularly suitable for detecting small molecules or analytes with a limited number of epitopes. One of the most well-known examples of heterogeneous immunoassays is the ELISA (enzyme-linked immunosorbent assay), which is routinely used in clinical laboratories for the detection and quantification of proteins [10].

In competitive immunoassays, commonly used for detecting small molecules, quantification relies on the competition between labeled and unlabeled analytes for a limited number of antibody binding sites, resulting in a signal inversely proportional to analyte concentration [11], [12].

Antigen-antibody pairs can exhibit different binding behaviors: a lower-affinity interaction may still generate a rapid signal due to a higher

association rate constant (k_{on}), while the dissociation rate constant (k_{off}) determines complex stability. In immunoassays, k_{on} influences binding speed and k_{off} affects signal retention, sensitivity, and overall performance. In rapid assays where equilibrium is not reached, these kinetics become critical, as fast-binding, lower-affinity ligands can temporarily occupy binding sites and alter signal response and calibration, requiring careful control of assay conditions [13], [14].

Immunoassay performance is defined by key parameters such as the limit of detection (LOD), dynamic range, precision, accuracy, and sensitivity, reflecting the assay's ability to detect low concentrations, quantify across a range, and produce reliable and responsive measurements [15].

The biological composition of clinical samples can significantly influence immunoassay performance. Proteins, lipids, and other endogenous components may interfere with antigen-antibody interactions or signal generation, reducing sensitivity and increasing the lower limit of detection (LOD) [16]. Among these, heterophilic antibodies (HAs) are particularly relevant. HAs are naturally occurring human antibodies capable of binding non-specifically to immunoassay antibodies, often causing false-positive or false-negative results [17]. Heterophilic antibodies (HAs) with both high and low affinity can interact with assay antibodies via Fc or Fab regions, forming bridges that mimic antigen binding or directly binding to the antigen, thereby perturbing assay signals. More importantly, HAs can directly bind to diagnostic IgG antibodies, either via Fc or Fab regions, forming non-specific bridges that mimic antigen binding and perturb assay signals. These interactions can reduce assay specificity, generate false-positive signals, or block true antigen-IgG interactions, leading to underestimation of analyte concentrations. Abundant plasma proteins, such as albumin and total immunoglobulins, can further influence IgG behavior by steric hindrance or by altering the local microenvironment around the IgG binding sites, indirectly modulating binding kinetics [18].

Additionally, lipids, hemoglobin, and small molecules such as biotin can interfere with assay performance.

In conventional laboratory methods, various strategies have been used to minimize matrix effects in samples. Among the most common approaches are sample dilution, which reduces the concentration of interfering substances, the use of blocking agents, which occupy non-specific binding sites and prevent false signals, and additional sample preparation or purification steps, such as centrifugation, filtration, or selective extraction, which remove interfering components from the sample [19].

Although effective, these strategies have practical limitations. Sample dilution can reduce assay sensitivity, blocking agents require careful optimization of reagents and conditions, and additional preparation steps increase the complexity of the analytical process, prolong analysis time, and often require specialized equipment such as centrifuges, filtration systems, or controlled incubators [19], [20].

Sample matrix in miniaturized systems for rapid diagnostics

In rapid diagnostic tests, reliable quantification in complex biological samples is challenging because matrix effects alter antibody–antigen binding kinetics, leading to inconsistent results within short analysis times [19], [21]. Therefore, low affinity interferent components can significantly impact microfluidic rapid diagnostics tests, while in laboratory tests these interferents rarely affect results due to long reagent incubation times [16]. Several strategies have been explored to address these challenges in microfluidic systems, however they mainly rely on plasma separation approaches, such as gravity separation, microstructures for blood cell retention, and spiral microchannels [22], [23]. However, sample interferents also exist in plasma, and even the low affinity binding ones affect microfluidic rapid tests [20], [24]. Reported

serum interference in sandwich immunoassays occurring in FEP (Fluorinated Ethylene Propylene) microcapillaries. This interference, probably due to low-affinity components could be minimized by antibody surface area optimization, microcapillary diameter and incubation times, suggesting that miniaturized rapid tests should be always developed in complex biological samples favoring their translation into clinical environments [20], [25].

Understanding these interactions enables the design of assays that perform reliably in complex biological samples. Molecular Dynamics simulations provide a way to study these interactions at atomic resolution, offering insights into binding mechanisms, structural stability, and the impact of complex sample environments on assay performance [26].

Molecular dynamics (MD) simulations have been widely applied to study antibody-antigen interactions, providing molecular-level insight into binding interfaces, conformational flexibility, and interaction energetics. Previous studies have investigated the structural stability of antibody-antigen complexes, identified key residues involved in binding, and analyzed intermolecular forces such as hydrogen bonding and hydrophobic interactions that govern molecular recognition [27]. Additionally, computational approaches have been used to explore binding affinity, conformational rearrangements, and the dynamics of immune complexes under physiological conditions. Coarse-grained methods further extend these capabilities by enabling the study of larger systems and longer timescales, which is particularly relevant for multivalent and complex biological environments [28]. However, most existing studies focus on idealized systems in buffer and do not account for the complexity of biological matrices or the role of low-affinity polyspecific antibodies in immunoassay interference.

Project objectives & methodology

The central problem of this project is the limited mechanistic understanding of how matrix components interfere with immunoassay

performance. This challenge arises from the complex composition of biological samples, the limited ability of experimental methods to isolate specific interference mechanisms, and the difficulty of translating buffer-optimized assays to real clinical conditions. Gaining molecular-level insight into these interactions is essential to improve assay reliability and enable accurate quantification in rapid diagnostic tests.

The main objective is to understand how naturally occurring polyspecific low-affinity antibodies mechanically interfere with immobilized immunoassays in microfluidic systems and affect binding kinetics, which is critical for quantitative rapid diagnostics.

The project will follow a computational approach based on molecular dynamics simulations, supported by literature-informed system design. Representative antigen-antibody complexes will be modeled alongside naturally occurring polyspecific low-affinity antibodies and other relevant interferents, such as albumin. Simulations will be performed using GROMACS, including system parameterization, solvation, energy minimization, and equilibration. Production runs will be conducted under physiological conditions, followed by analysis of binding stability, interaction energies, and conformational changes. The influence of these interferents on binding kinetics and affinity will be evaluated to identify the molecular mechanisms driving matrix effects and to inform improved immunoassay design.

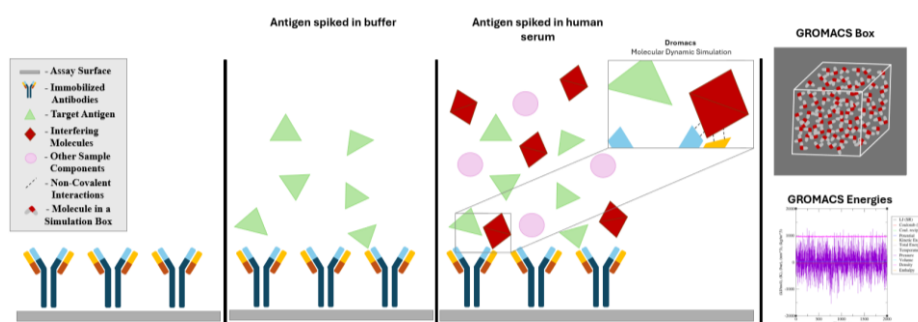


Figure 1 – Project problem and methodology – graphical abstract.

References:

1. Anderson, N. L. & Anderson, N. G. The Human Plasma Proteome. *Molecular & Cellular Proteomics* **1**, 845–867 (2002).
2. Zhang, J. *et al.* Recent advances in biomarkers for predicting the efficacy of immunotherapy in non-small cell lung cancer. *Front. Immunol.* **16**, (2025).
3. Mojebi, A. *et al.* Clinical impact of rapid molecular diagnostic tests in patients presenting with viral respiratory symptoms: A systematic literature review. *PLoS One* **19**, e0303560 (2024).
4. McNerney, M. P. *et al.* Point-of-care biomarker quantification enabled by sample-specific calibration. *Sci. Adv.* **5**, eaax4473 (2019).
5. Pecoraro, V., Banfi, G., Germagnoli, L. & Trenti, T. A systematic evaluation of immunoassay point-of-care testing to define impact on patients' outcomes. *Ann. Clin. Biochem.* **54**, 420–431 (2017).
6. Whitesides, G. M. The origins and the future of microfluidics. *Nature* **2006 442:7101** **442**, 368–373 (2006).
7. Sackmann, E. K., Fulton, A. L. & Beebe, D. J. The present and future role of microfluidics in biomedical research. *Nature* **2014 507:7491** **507**, 181–189 (2014).
8. Liu, J., Xu, Z., Shan, Y. & Huang, X. Applications of microcapillary films in bioanalytical techniques. *Analyst* **146**, 1529–1537 (2021).
9. Daniels, E., Mustafa, Y. L., Herdes, C. & Leese, H. S. Optimization of Cortisol-Selective Molecularly Imprinted Polymers Enabled by Molecular Dynamics Simulations. *ACS Appl. Bio Mater.* **4**, 7243–7253 (2021).

10. Hayrapetyan, H., Tran, T., Tellez-Corrales, E. & Madiraju, C. Enzyme-Linked Immunosorbent Assay: Types and Applications. *Methods Mol. Biol.* **2612**, 1–17 (2023).
11. Zhang, B. *et al.* Improved Sensitivity and Wide Range Detection of Small Analytes Using a Two-Antigen-Combined Competitive Immunoassay. *ACS Omega* **7**, 48121–48129 (2022).
12. Murphy, B. M., He, X., Dandy, D. & Henry, C. S. Competitive Immunoassays for Simultaneous Detection of Metabolites and Proteins Using Micromosaic Patterning. *Anal. Chem.* **80**, 444 (2007).
13. Dinh, T. L., Ngan, K. C., Shoemaker, C. B. & Walt, D. R. Using Antigen–antibody Binding Kinetic Parameters to Understand Single-Molecule Array Immunoassay Performance. *Anal. Chem.* **88**, 11335–11339 (2016).
14. Motulsky, H. & Christopoulos, A. Analyzing Kinetic Binding Data. *Fitting Models to Biological Data Using Linear and Nonlinear Regression* 233–244 (2021)
doi:10.1093/oso/9780195171792.003.0039.
15. Tate, J. & Ward, G. Interferences in Immunoassay. *Clin. Biochem. Rev.* **25**, 105 (2004).
16. Bolstad, N., Warren, D. J. & Nustad, K. Heterophilic antibody interference in immunometric assays. *Best Pract. Res. Clin. Endocrinol. Metab.* **27**, 647–661 (2013).
17. Levinson, S. S. Antibody multispecificity in immunoassay interference. *Clin. Biochem.* **25**, 77–87 (1992).
18. Warade, J. Retrospective Approach to Evaluate Interferences in Immunoassay. *EJIFCC* **28**, 224 (2017).

19. Bolstad, N. *et al.* Heterophilic antibody interference in commercial immunoassays; a screening study using paired native and pre-blocked sera. *Clin. Chem. Lab. Med.* **49**, 2001–2006 (2011).
20. Rosenberg-Hasson, Y., Hansmann, L., Liedtke, M., Herschmann, I. & Maecker, H. T. Effects of serum and plasma matrices on multiplex immunoassays. *Immunol. Res.* **58**, 224 (2014).
21. Barbosa, A. I., Edwards, A. D. & Reis, N. M. Antibody Surface Coverage Drives Matrix Interference in Microfluidic Capillary Immunoassays. *ACS Sens.* **6**, 2682 (2021).
22. Richards, D. A. & deMello, A. J. The Matrix Reiterated. *ACS Sens.* **8**, 3986–3987 (2023).
23. Whitesides, G. M. The origins and the future of microfluidics. *Nature* **442**, 7101 **442**, 368–373 (2006).
24. Yang, S. M., Lv, S., Zhang, W. & Cui, Y. Microfluidic Point-of-Care (POC) Devices in Early Diagnosis: A Review of Opportunities and Challenges. *Sensors (Basel)* **22**, 1620 (2022).
25. Tighe, P. J., Ryder, R. R., Todd, I. & Fairclough, L. C. ELISA in the multiplex era: Potentials and pitfalls. *Proteomics Clin. Appl.* **9**, 406 (2015).
26. Mackey, D., Kelly, E., Nooney, R. & O’Kennedy, R. Direct immunoassays and their performance - theoretical modelling of the effects of antibody orientation and associated kinetics. *Integr. Biol. (Camb)*. **10**, 598–604 (2018).
27. Xu, H., Lu, J. R. & Williams, D. E. Effect of surface packing density of interfacially adsorbed monoclonal antibody on the binding of hormonal antigen human chorionic gonadotrophin. *J. Phys. Chem. B* **110**, 1907–1914 (2006).

28. Karbalaee, Z., Rezayan, A. H., Taheri, R. A., Mirhosseini, S. A. & Barshan-Tashnizi, M. Enhancing Shiga toxin detection using surface plasmon resonance a study of antibody immobilization strategies. *Scientific Reports* 2025 15:1 **15**, 17024- (2025).
29. Hollingsworth, S. A. & Dror, R. O. Molecular dynamics simulation for all. *Neuron* **99**, 1129 (2018).